INCH-POUND
MIL-PRF-8805/109D
19 November 2018
SUPERSEDING
MIL-PRF-8805/109C
12 December 2006

PERFORMANCE SPECIFICATION SHEET

SWITCH, SENSITIVE, SPDT, UNSEALED, 7 AMPERES, 0.156 WIDTH BASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this document and MIL-PRF-8805.

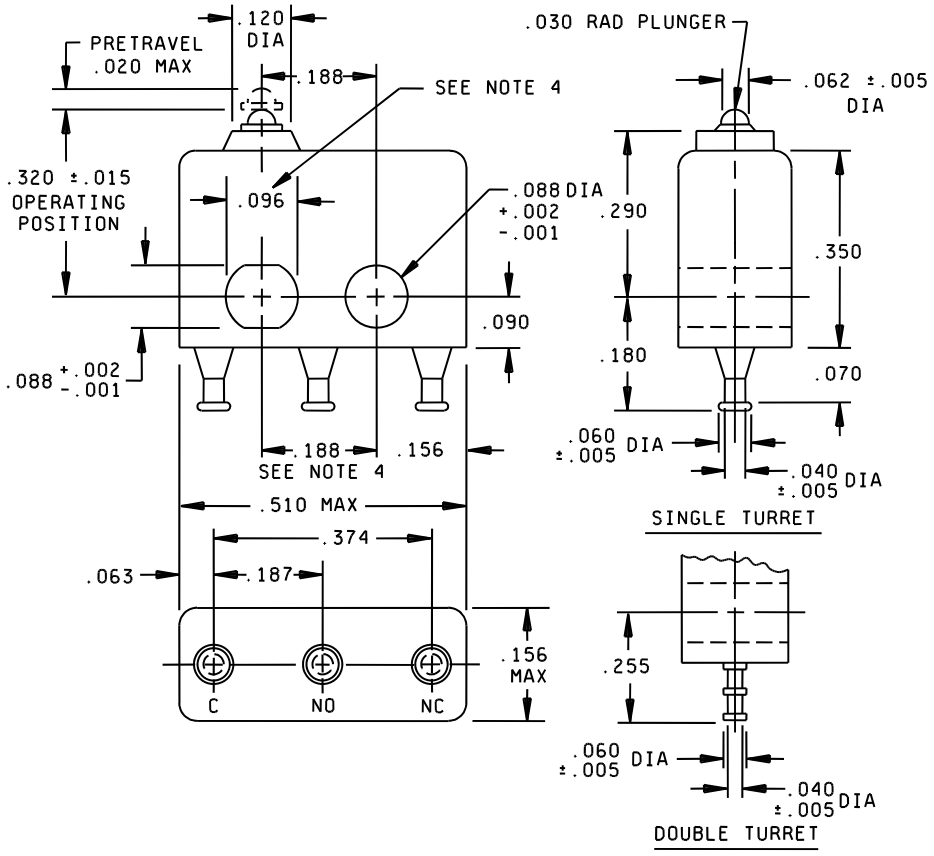


FIGURE 1. Configuration and dimensions.

AMSC N/A

FSC 5930

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:22Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	.03	.020	.51	.063	1.60	.096	2.44	.188	4.78	.374	9.50
.002	.05	.030	.76	.070	1.78	.120	3.05	.255	6.48	.510	12.95
.005	.13	.040	1.02	.087	2.21	.156	3.96	.290	7.37		
.010	.25	.060	1.52	.088	2.24	.180	4.57	.320	8.13		
.015	.38	.062	1.57	.090	2.29	.187	4.75	.350	8.89		

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specifies, tolerances are ± .010 (.25 mm).
3. Metric equivalents are given for general information only.
4. Mounting holes will accept pins or screws of .087 maximum diameter on .188 ± .002 centers.
5. Double turret terminals shall accept two AN-20 wires or equivalent.

FIGURE 1. Configuration and dimensions.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Enclosure design: 1 (unsealed)

Temperature characteristics: 1 (- 55°C to + 85°C).

Shock type: M (100 g).

Vibration grade: 1 (10 to 500 Hz).

Weight: .003 pounds maximum.

Operating characteristics:

Actuating force: 5 ounces maximum.

Release force: 1 ounce minimum.

Movement differential: .005 inch maximum.

Pretravel: .020 inch maximum.

Overtravel: .004 inch minimum.

Terminals: Terminal finish may be .000030 inch minimum to .000100 inch maximum thick gold plate or equivalent.

Terminal strength: 5 pounds, all directions.

Mechanical endurance: 50,000 cycles at .004 + .001, -.000 inch overtravel.

Dielectric withstanding voltage:

Sea level: 1050 V rms.

Altitude (50,000 ft); 400 V rms.

Electrical endurance: 25,000 cycles.

Electrical ratings: See tables I and II.

Part or Identifying Number (PIN): See table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:22Z
Check the source to verify that this is the current version before use.

Low level circuit: 25,000 cycles, applicable to switches with gold contacts (PIN's M8805/109-03 and M8805/109-04).

Intermediate current: 25,000 cycles.

Qualification:

Group submission: See table IV.

TABLE I. Electrical ratings: silver contacts.

Load	Sea level		50,000 feet
	28 V dc	115 V ac, 60 Hz	28 V dc
	Amperes	Amperes	Amperes
Resistive	7.0	7.0	4.0
Inductive	4.0	7.0	2.5
Lamp	2.5	2.0	2.5

TABLE I. Electrical ratings: gold contacts.

Load	Sea level 28 V dc	50, 000 feet 28 V dc
	amperes	amperes
Resistive	1	1
Inductive	.5	.5

TABLE III. PIN's and characteristics.

PIN's	Contact material	Terminals
M8805/109-01	Silver	Single turret
M8805/109-02	Silver	Double turret
M8805/109-03	Gold	Single turret
M8805/109-04	Gold	Double turret

TABLE IV. Qualification inspection.

PIN's	Inspection	Extent of approval
M8805/109-02	Qualification inspection table of MIL-PRF-8805	ALL
M8805/109-03 (10 sample units)	Groups I, VII, and X of qualification inspection table	

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:22Z
Check the source to verify that this is the current version before use.

Referenced documents:
MIL-PRF-8805

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Navy - EC
DLA - CC

Preparing activity
DLA – CC

(Project 5930-2018-059)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:22Z
Check the source to verify that this is the current version before use.